KARL FITENI

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POSITIONS

Postdoctoral Researcher *Oct 2024 – present*
Como Lake centre for AstroPhysics (CLAP), Università dell'Insubria, Italy. ERC Starting Grant "GalFlow" (PI M. C. Sormani)

Affiliate Researcher *Jan 2024 – present*
Institute of Space Sciences & Astronomy (ISSA), University of Malta

Research visits: Ghent University, Belgium (1 month, 2020); Jeremiah Horrocks Institute, UK (2019, COST-funded; 2022)

EDUCATION

PhD, Astrophysics, University of Malta (ISSA) *2019 – 2024*
Thesis: *Secular Evolution in the Milky Way Galaxy.* Supervisors: Prof. J. Caruana (principal supervisor), Prof. V. P. Debattista (co-supervisor). Awarded 7 November 2024.

MSc, Astrophysics, University of Malta (ISSA) *2017 – 2018*

BSc (Hons), Mathematics and Physics, University of Malta *2013 – 2017*

AWARDS AND GRANTS

2025 **Outstanding PhD Thesis Award (STEM)**, University of Malta – one award across all STEM doctoral theses university-wide
2020 Tertiary Education Scholarship Scheme (TESS), Ministry of Education, Malta
2019 COST Short-Term Scientific Mission grant (CA18104, *Revealing the Milky Way with Gaia*)
2018 COST Short-Term Scientific Mission grant (BigSkyEarth, GPU-based data analytics)
2017 Dean's Award (Certificate of Merit) for the final-year dissertation, University of Malta

PUBLICATIONS

14 papers, 5 as first author (all published): 12 accepted, 2 submitted.
Full record: NASA ADS

First-author papers

1. **K. Fiteni**, M. C. Sormani, V. P. Debattista, F. Nogueras-Lara, R. Schödel, J. L. Sanders, M. Schultheis, X. Li, A. Vasini, Z.-X. Feng, M. Donati. *Kinematic hints of a nuclear bar in the Milky Way.* A&A, 714, A181 (2026).

2. **K. Fiteni**, X. Li, M. C. Sormani, V. P. Debattista, A. Vasini, F. Nogueras-Lara, J. L. Sanders, N. Deg, M. Schultheis, M. Donati, Z.-X. Feng. *Kinematic diagnostics for non-axisymmetry in the Milky Way's nuclear stellar disc.* A&A, 709, A40 (2026).

3. **K. Fiteni**, S. R. Anderson, V. P. Debattista, J. Caruana, J. A. S. Amarante, S. Gough-Kelly, L. Eyer, L. Beraldo e Silva, T. Khachaturyants, V. Cuomo. *The edge of the Milky Way's star-forming disc: evidence from a 'U-shaped' stellar age profile.* A&A, 708, A252 (2026). Covered by *Sky & Telescope*, Space.com and *Universe Today*.

4. **K. Fiteni**, S. De Rijcke, V. P. Debattista, J. Caruana. *The role of density breaks in driving spiral structure in disc galaxies.* MNRAS, 529, 4879 (2024).

5. **K. Fiteni**, J. Caruana, J. A. S. Amarante, V. P. Debattista, L. Beraldo e Silva. *The relative efficiencies of bars and clumps in driving disc stars to retrograde motion.* MNRAS, 503, 1418 (2021).

Co-authored papers

1. J. A. S. Amarante, C. F. P. Laporte, V. P. Debattista, et al. (13 authors, incl. **K. Fiteni**). *GASTRO Library. II. Exploring chemical bimodalities in disk galaxies with GSE-like mergers and massive star-forming clumps.* ApJ, 1004, 242 (2026).
   My contribution: the code that detects the clumps and measures clumpiness (Fig. 1).

2. S. N. Longmore et al. (ACES Collaboration, 98 authors, incl. **K. Fiteni**). *ALMA Central Molecular Zone Exploration Survey (ACES) – I. Overview.* MNRAS, 552, stag377 (2026).
   My contribution: scientific discussion and manuscript feedback, pre- and post-review.

3. Y. Sofue et al. (26 authors, incl. **K. Fiteni**). *Rotation and stability of the circumnuclear gas disk in the Galactic Center potential by the ALMA CMZ Exploration Survey (ACES).* PASJ, 78, 536 (2026).
   My contribution: scientific discussion and manuscript feedback, pre- and post-review.

4. F. Nogueras-Lara, A. T. Barnes, J. D. Henshaw, **K. Fiteni**, et al. (28 authors). *Unveiling the 3D structure of the central molecular zone from stellar kinematics and photometry: the 50 and 20 km s^{-1} clouds.* A&A, 706, A18 (2026).
   My contribution: the theoretical analysis (Fig. 7): the proper-motion–distance profile from an N-body realisation of the nuclear-stellar-disc model, used to place the clouds along the line of sight.

5. M. Schultheis, L. Serrano, B. Thorsbro, F. Nogueras-Lara, A. Feldmeier-Krause, G. Nandakumar, **K. Fiteni**, M. C. Sormani, N. Ryde. *Chemical analysis of the Milky Way's*

nuclear star cluster: evidence for a metallicity gradient. A&A, 705, A235 (2026).
My contribution: scientific discussion and manuscript feedback, pre- and post-review.

6. S. Joharle, F. Nogueras-Lara, **K. Fiteni**. *Kinematic and extinction analysis of a potential spiral arm beyond the Galactic bar.* A&A, 705, A131 (2026).
My contribution: the N-body simulation analysis (Sect. 4.3), confirming that the observed proper-motion offset arises from stars at different distances along the line of sight.

7. Z.-X. Feng, M. C. Sormani, R. G. Tress, et al. (19 authors, incl. **K. Fiteni**). *Simulations of gas inflow in the Milky Way I. Stellar-feedback-regulated transport from the Central Molecular Zone to the circumnuclear disk.* Submitted, arXiv:2605.16192 (2026).
My contribution: scientific discussion and manuscript feedback, pre- and post-review.

8. A. Vasini, M. C. Sormani, J. L. Sanders, et al. (15 authors, incl. **K. Fiteni**). *A novel approach to 3D dust mapping of the Central Molecular Zone.* A&A, submitted, arXiv:2609.10782 (2026).
My contribution: scientific discussion and manuscript feedback, pre- and post-review.

9. M. Du, V. P. Debattista, L. C. Ho, et al. (12 authors, incl. **K. Fiteni**). *The formation of compact elliptical galaxies in the vicinity of a massive galaxy: the role of ram-pressure confinement.* ApJ, 875, 58 (2019).
My contribution: the code to trace gas particles between snapshots (Figs. 9–10).

COLLABORATIONS

KMOS VVVX-GalCen ESO Public Survey
Collaborator, via co-PI F. Nogueras-Lara (programme 116.29EF; PI M. Gómez). The survey's Galactic-centre component targets ∼40 000 nuclear-stellar-disc stars. I lead the nuclear-bar analysis of these data, with first access to the Galactic-centre pilot (∼3 000 stars).

JWST/NIRCam Legacy Survey of the Galactic Center *2026 – present*
Team member, GO 10678 (143 h Cycle 5 Treasury; PI R. Schödel).

ALMA CMZ Exploration Survey (ACES) *2024 – present*
Member, ALMA Large Program 2021.1.00172.L (∼100 members). Stellar dynamics of the nuclear stellar disc; co-author of the survey overview paper.

GASTRO Library
Member. N-body+SPH simulations of *Gaia*-Sausage-Enceladus-like mergers and their imprint on Milky Way-like discs and haloes.

TALKS

Invited

2026 Workshop *Galactic Nuclei Astrophysical Phenomena*, Sexten, Italy – "A nuclear bar in the Milky Way? Kinematic hints from the Galactic Centre"

2026 Invited seminar (online), research group of Dr I. Minchev, Leibniz Institute for Astrophysics Potsdam (AIP) – "A nuclear bar in the Milky Way? Kinematic hints from the Galactic Centre"

2025 JASMINE European Collaboration pre-kick-off meeting – "Kinematic diagnostics for non-axisymmetry in the Milky Way's nuclear stellar disc"

Contributed

2026 *Galactic Bars, Bulges and Disks* (Astro Lipari 2026), Lipari, Italy – "The edge of the Milky Way's star-forming disc"

2026 IAU Symposium 405 *Traversing the Galactic Center in Space and Time*, Brno, Czech Republic – "A nuclear bar in the Milky Way? Kinematic hints from the Galactic Centre"

2024 European Astronomical Society Annual Meeting, Padova, Italy – "The role of stellar density breaks in driving spiral structure in disc galaxies"

2021 N-Body Shop Excellence Conference (online) – "The relative efficiencies of bars and clumps in driving disc stars to heated orbits"

MENTORING AND SUPERVISION

Current

Z. Galicka (2026–present) – nuclear boxy/peanut bulges in simulations of double-barred galaxies (began as a summer project)

Guiding undergraduate students (dissertations), University of Malta

D. Attard (2022–23) – MHD simulations of bars, spirals and the star-forming disc
L.-M. Aquilina (2022–23) – Do bending waves in the gas cause the Gould Belt?
F. Borg (2022–23) – Orbital trapping by bars
M. Micelli (2022–23) – The cosmic evolution of boxy/peanut bulges
M. Aquilina (2021–23) – Spirals in the Milky Way: predicted tests for *Gaia*
F. Sant (2021–22) – Bars, spirals and the radial extent of a star-forming disc

Summer research projects in galactic dynamics: D. Attard, F. Borg, L.-M. Aquilina (2021)

ORGANISATION AND SERVICE

2026 Co-organiser and LOC, *Galactic Bars, Bulges and Disks* (Astro Lipari 2026), Lipari, Italy – astrolipari2026.com

2025 Co-organiser, Inaugural Workshop of the Como Lake centre for AstroPhysics, Sormano Observatory, Italy

2017 LOC, *Gas in Galaxies*, University of Malta, Valletta

2026 Referee, *The Astrophysical Journal Letters*

2026 Referee, *Astronomy & Astrophysics*

TEACHING

University of Malta, Department of Physics (2019 – 2022): tutor for PHY3232 Nuclear and Particle Physics; laboratory demonstrator for PHY3111 Advanced Experimental Physics, PHY2230 Experimental Physics and PHY1105 Fundamentals of Experimental Physics.

PUBLIC ENGAGEMENT

University of Malta outreach (from my undergraduate studies through my PhD): regular volunteer at the University's public outreach events, such as *Science in the City*, Malta's annual science and arts festival, and school visits promoting STEM subjects and astronomy to secondary-school students.

Press: my study of the edge of the Milky Way's star-forming disc was covered by *Sky & Telescope*, Space.com and *Universe Today* (2026).

Public Relations Officer, Astronomical Society of Malta (2019 – 2021): ran the Society's social media, significantly increasing engagement and contributing to a rise in active membership; organised observing nights, public talks and online events; live interview on *TVAM* (TVM, Malta's national broadcaster).

Popular article: "How bright?", *THINK* magazine, University of Malta, issue 19 (2017) – on my undergraduate dissertation measuring variable stars from Malta.

TECHNICAL SKILLS

Dynamics and simulations: analysis of N-body+SPH simulations; linear stability analysis of disc models (PyStab, C/C++); orbit integration and potentials with Agama and galpy.

Data analysis: Python (pynbody, Agama, galpy, Astropy, SuperFreq, emcee, scikit-learn, pandas, SciPy); *Gaia*, APOGEE, LAMOST, KMOS and VIRAC2 data; Gaussian-mixture modelling and bootstrap/MCMC inference.

ULTRA-DISTANCE RUNNING

2026 Lake Como Ultra Trail (120 km)
2026 Lavaredo Ultra Trail (80 km)
2026 Ultra Malta Trail (50 km)
2025 Jurassic Miles Endurance Run (160 km)
2025 Endure Backyard Ultra (147 km)